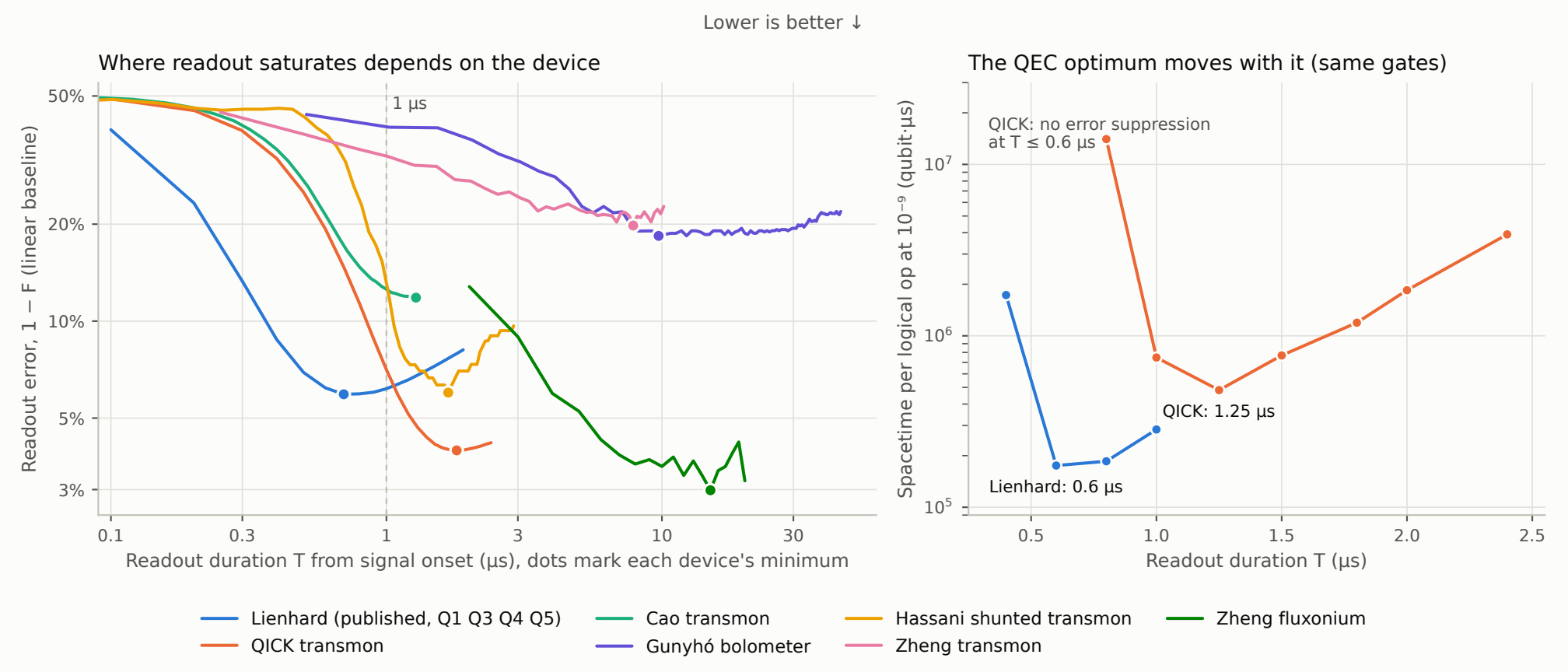
Lower is better ↓
Where readout saturates depends on the device
Readout error, 1 − F (linear baseline)
50%
20%
10%
5%
3%
1 µs
0.1
0.3
1
3
10
30
Readout duration T from signal onset (µs), dots mark each device's minimum
The QEC optimum moves with it (same gates)
Spacetime per logical op at 10^{-9} (qubit·µs)
10^7
10^6
10^5
QICK: no error suppression
at T ≤ 0.6 µs
QICK: 1.25 µs
Lienhard: 0.6 µs
0.5
1.0
1.5
2.0
2.5
Readout duration T (µs)
Lienhard (published, Q1 Q3 Q4 Q5)
QICK transmon
Cao transmon
Gunyhó bolometer
Hassani shunted transmon
Zheng transmon
Zheng fluxonium